Assignment of Deep Learning for Medical Imaging

BUSI Dataset Classification

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Abstract

Breast cancer detection using medical imaging is an important application of machine learning. In this project, a Convolutional Neural Network (CNN) is used to classify breast ultrasound images from the BUSI dataset into three categories: benign, malignant, and normal. A major challenge in this dataset is class imbalance, where some classes contain significantly more samples than others. To study the effect of this imbalance, a baseline CNN model is first trained without any special handling. Then two techniques are applied to address the imbalance: Synthetic Minority Oversampling Technique (SMOTE) and class weighting during training. The performance of these approaches is compared to understand how imbalance handling methods affect classification results. This work demonstrates the importance of handling imbalanced data when building reliable deep learning models for medical image classification.

1 Introduction

Breast cancer is one of the most common diseases affecting women worldwide, and early diagnosis is essential for improving survival rates. Medical imaging techniques such as ultrasound are widely used for detecting abnormalities in breast tissue. In recent years, machine learning and deep learning techniques have been applied to automate the analysis of medical images. In this work, a Convolutional Neural Network (CNN) is used to classify breast ultrasound images. The study also explores the impact of class imbalance and evaluates techniques such as SMOTE and class weighting to improve classification performance.

2 Dataset Description

The dataset used in this study is the Breast Ultrasound Images Dataset (BUSI). It contains ultrasound images categorized into three classes: benign, malignant, and normal. The dataset contains 780 ultrasound images in total, including 437 benign, 210 malignant, and 133 normal cases. Each image also includes a corresponding segmentation mask; however, only the original images are used for classification. The dataset is imbalanced, with a larger number of benign images compared to malignant and normal images. All images are resized to a fixed resolution and normalized before being used for training the model.

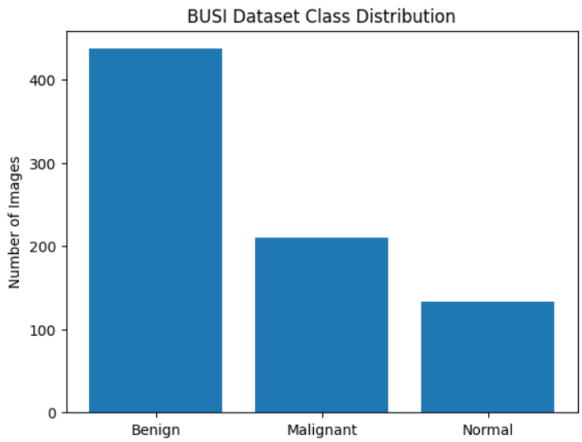

Figure 1: Class distribution in the BUSI dataset

3 Methodology

The methodology used in this project consists of several stages including data preprocessing, model design, and handling of class imbalance. The goal is to evaluate how different imbalance handling techniques affect the performance of a Convolutional Neural Network (CNN) for breast ultrasound image classification.

3.1 Data Preprocessing

The Breast Ultrasound Images Dataset (BUSI) contains ultrasound images belonging to three classes: benign, malignant, and normal. Each image also includes a segmentation mask; however, the masks are not used in this classification task. Only the original images are selected.

All images are resized to a fixed dimension of 128×128 pixels to ensure uniform input to the neural network.

Pixel values are normalized to the range $[0, 1]$ by dividing by 255. The dataset is then split into training and testing sets using stratified sampling to preserve the class distribution.

3.2 Baseline CNN Model

A Convolutional Neural Network (CNN) is used to automatically extract spatial features from the ultrasound images. The architecture consists of multiple convolutional layers followed by max-pooling layers to capture important patterns and reduce spatial dimensions. These layers are followed by a flattening layer and fully connected dense layers for classification.

The final layer uses a softmax activation function to output probabilities for the three classes. The model is trained using the Adam optimizer with categorical cross-entropy as the loss function.

3.3 Handling Class Imbalance

Since the dataset contains an unequal number of samples in each class, imbalance handling techniques are applied to improve model performance.

SMOTE (Synthetic Minority Oversampling Technique): SMOTE is used to generate synthetic samples for the minority classes in the training dataset. This technique creates new data points by interpolating between existing minority samples, which helps balance the dataset and allows the model to learn better representations for underrepresented classes.

Class Weighting: In this approach, different weights are assigned to each class during training. Classes with fewer samples are given higher weights so that misclassification of minority classes is penalized more heavily. This encourages the model to pay greater attention to minority classes during learning.

3.4 Model Evaluation

The models are evaluated on the test dataset using standard classification metrics including accuracy, precision, recall, and F1-score. These metrics provide a more reliable evaluation for imbalanced datasets compared to accuracy alone. The results from the baseline CNN, CNN with SMOTE, and CNN with class weighting are compared to analyze the effectiveness of the imbalance handling methods.

3.5 CNN Architecture

The classification model is implemented using a Convolutional Neural Network (CNN). The network takes an input image of size $128 \times 128 \times 3$. Convolutional layers are used to extract spatial features from the images, while max-pooling layers reduce the spatial dimensions and help the network focus on important patterns.

After feature extraction, the feature maps are flattened and passed through fully connected layers for classification. A dropout layer is included to reduce overfitting during training. The final layer uses a softmax activation function to produce probability scores for the three classes: benign, malignant, and normal.

Layer	Configuration	Output Size
Input	Image	$128 \times 128 \times 3$
Conv2D	32 filters, 3×3, ReLU	$126 \times 126 \times 32$
MaxPool	2×2	$63 \times 63 \times 32$
Conv2D	64 filters, 3×3, ReLU	$61 \times 61 \times 64$
MaxPool	2×2	$30 \times 30 \times 64$
Conv2D	128 filters, 3×3, ReLU	$28 \times 28 \times 128$
MaxPool	2×2	$14 \times 14 \times 128$
Flatten	Feature vector	25088
Dense	128 units, ReLU	128
Dropout	Rate 0.5	128
Output	3 units, Softmax	3

Table 1: CNN architecture

4 Results

The performance of three models was evaluated: a baseline CNN model, CNN trained with SMOTE, and CNN trained using class weights. The models were compared using accuracy, precision, recall, and F1-score. These metrics are particularly useful for imbalanced datasets because they provide better insight into class-wise performance.

The quantitative comparison of the models is shown in Table 2, while Figure 2 illustrates the performance differences between the models.

The baseline CNN achieved an overall accuracy of 0.72. While the model performed reasonably well for the majority class, it showed poor recall for the minority class (class 2), indicating difficulty in detecting underrepresented samples.

When SMOTE was applied, the dataset became more balanced during training, leading to improved overall performance. The CNN trained with SMOTE achieved the highest accuracy of 0.77 and significantly improved the recall and F1-score for the minority class compared to the baseline model. This indicates that synthetic oversampling helped the model learn better representations of the minority class.

The CNN model trained with class weighting also improved the detection of minority classes. Although its overall accuracy (0.74) was slightly lower than the SMOTE model, the recall for class 2 increased to 0.74, demonstrating that the model gave more importance to the underrepresented class during training.

Overall, both imbalance handling techniques improved the classification performance compared to the baseline CNN. Among the tested methods, SMOTE provided the best overall accuracy, while class weighting helped achieve more balanced recall across the classes.

5 Conclusion

In this project, a Convolutional Neural Network (CNN) was implemented to classify breast ultrasound images

Model	Accuracy	Precision	Recall	F1-score
Baseline CNN	0.72	0.74	0.72	0.69
CNN + SMOTE	0.77	0.77	0.77	0.76
CNN + Class Weight	0.74	0.75	0.74	0.74

Table 2: Performance comparison of different models

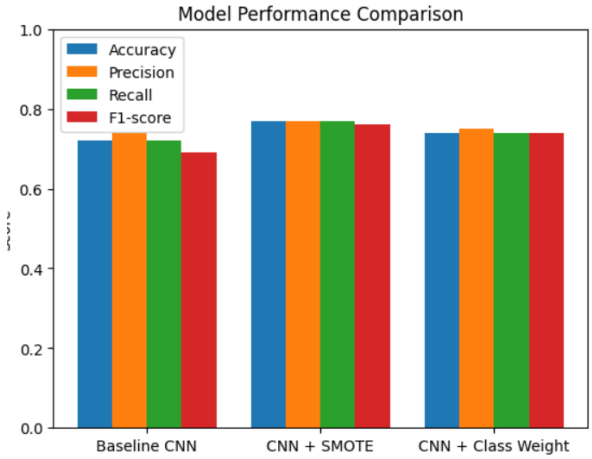

Figure 2: Comparison of performance metrics for different models

from the BUSI dataset into benign, malignant, and normal categories. A baseline CNN model was first trained to establish a reference performance. Since the dataset is imbalanced, two techniques were explored to improve the model's ability to detect minority classes: SMOTE and class weighting.

The experimental results show that handling class imbalance improves the classification performance compared to the baseline model. The CNN trained with SMOTE achieved the highest overall accuracy and F1-score, indicating better generalization across classes. The class weight approach also improved the detection of minority classes by assigning higher importance to underrepresented samples during training.

Overall, the study demonstrates that addressing class imbalance is important for building reliable deep learning models in medical image classification tasks. Future work could explore more advanced techniques such as transfer learning or data augmentation to further improve model performance.

References

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